minimum of two line segments are needed before a polygon can be created by the third closing segment.)

The Close option is used to join the current point to the start point of the first line when two or more continuous lines are drawn. If you specify the endpoint by using the mouse, then click at the start point of the first line or enter C at the Command prompt, as given in the Command prompt below.

Command: LINE
Specify first point: Pick the first point.
Specify next point or [Undo]: Pick the second point.
Specify next point or [Undo]: Pick the third point.
Specify next point or [Close/Undo]: C (The fourth point joins with the first point).

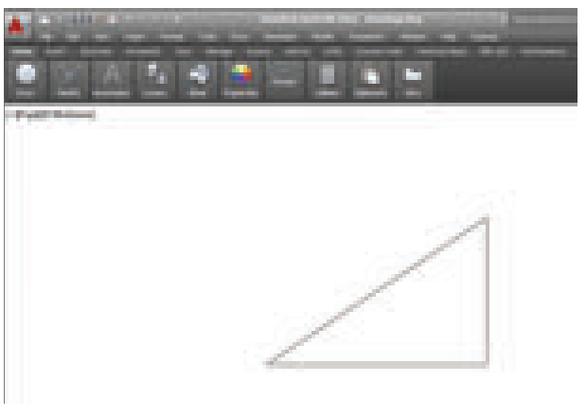


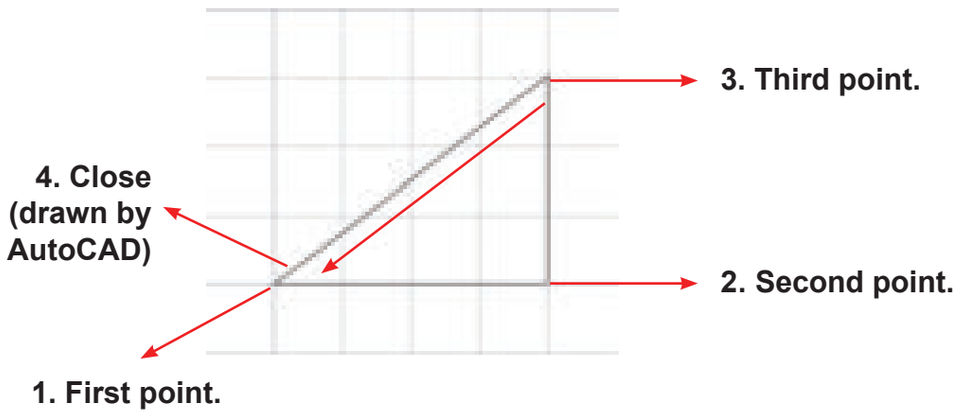
Fig. 6.55 Drawing lines using Close option

The drawing seen on the screen is a triangle of arbitrary size. To prepare the correct dimensional engineering drawings, the Coordinate systems is used.

6.7.2 COORDINATE SYSTEMS

In AutoCAD, the location of a point is specified in terms of Cartesian coordinates. In this system, each point in a plane is specified by a pair of numerical coordinates. To specify a point in a plane, take two mutually perpendicular lines as references. The horizontal line is called the X axis, and the vertical line is called the Y axis. The X and Y axes divide the XY plane into four parts, generally known as quadrants. The point of intersection of these two axes is called the origin and the plane is called the XY plane.

The origin has the coordinate values of X = 0, Y = 0. The origin is taken as the reference for locating a point on the XY plane. Now, to locate a point, say P, draw a vertical line intersecting the X axis. The horizontal distance between the origin and the intersection point will be called the X coordinate of P. It will be denoted as P(x). The X coordinate specifies how far the point is to the left or right from the origin along the X axis. Now, draw a horizontal line intersecting the Y axis.



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